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ORIGINAL ARTICLE *Embryology*

Inter- and intra-observer variability of time-lapse annotations

Linda Sundvall[1,2,*], Hans Jakob Ingerslev[1,2], Ulla Breth Knudsen[1,2], and Kirstine Kirkegaard[1,2]

[1]Fertility Clinic, Aarhus University Hospital, Brendstrupgaardsvej 100, 8200 Aarhus, Denmark [2]Health, Aarhus University, Vennelyst Boulevard 9, Aarhus C DK-8000, Denmark

*Correspondence address. Tel: +45-61-66-34-24; E-mail: linda.karin.sundvall@post.au.dk

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STUDY QUESTION: How consistent is the time-lapse annotation of dynamic and static morphologic parameters of embryo development, within and between observers?

SUMMARY ANSWER: The assessment of dynamic parameters is characterized by almost perfect agreement within and between observers.

WHAT IS KNOWN ALREADY: The commonly employed method used to assess embryos in IVF treatments is based on static evaluation of morphology in a microscope, but this is limited by substantial intra- and inter-observer variation. Time-lapse imaging has been proposed as a method to refine embryo selection by adding new dynamic predictors of viability to the assessment. Yet, there are no data regarding the consistency of estimates of the time-lapse parameters.

STUDY DESIGN, SIZE, DURATION: Infertile patients were recruited at the Fertility Clinic, Arhus University Hospital from February 2011 to June 2012. All embryos were cultured for 6 days in a time-lapse incubator (EmbryoScope™). Automated image recording was performed every 20 min.

PARTICIPANTS/MATERIALS, SETTING, METHODS: In total, 158 fertilized embryos from 20 different patients were annotated. Three observers made independent annotations on time-lapse recordings. One observer performed the assessment twice. Twenty-five parameters were annotated and the inter- and intra-observer agreement was assessed by calculating intra-class correlation coefficients (ICCs).

MAIN RESULTS AND THE ROLE OF CHANCE: Extremely close agreement (ICC 0.99) was found for dynamic parameters including the timing of the following: pronuclei breakdown, completion of blastocyst hatching and the appearance and disappearance of the first nucleus after the first division. Observations of cleavage divisions were strongly correlated (ICC > 0.8), indicating close agreement. Measurements of the static morphologic parameters, i.e. multi-nucleation and evenness of blastomeres at 2-cell stage showed fair-to-moderate agreement (ICC ≤ 0.5).

LIMITATIONS, REASON FOR CAUTION: The study was conducted at a single clinic. Only embryos with a good prognosis were included. The influence of training sessions was not measured.

WIDER IMPLICATIONS OF THE FINDINGS: Consistency is crucial to the validity of embryo scoring and selection. All of the time-lapse parameters suggested by the literature showed in our study high intra- and inter-observer correlation, thus validating the precision of time-lapse annotations. This provides the basis for further investigation of embryo assessment and selection by time-lapse imaging in prospective trials.

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Introduction

Time-lapse analysis is emerging as a method for assessing embryo quality in IVF treatments. Several retrospective cohort studies have correlated the timing and duration of events to viability and pregnancy (Wong *et al.*, 2010; Meseguer *et al.*, 2011; Azzarello *et al.*, 2012; Cruz *et al.*, 2012; Dal Canto *et al.*, 2012; Hashimoto *et al.*, 2012; Hlinka *et al.*, 2012), thus introducing potential new dynamic parameters of embryo quality. Time lapse has therefore been proposed as a method to improve embryo selection, which would be expected to increase clinical pregnancy rates, as recently reported by (Meseguer et al., 2012) in a study based on historical data. Nevertheless, the most predictive parameters remain to be validated in larger prospective randomized trials in order to prove their clinical relevance. The validity of embryo assessment and selection by time-lapse imaging will only be established, however, if it can be shown to be accurate and reproducible. A crucial step in evaluating such parameters is therefore to examine the consistency of the assessment. In contrast to the traditional static evaluation, time-lapse analysis enables one or more observers to make repeated observations on the same embryo. This allows for a potentially improved control of the quality of the assessment and selection.

Consistency and accuracy depend on the interplay of variability within and between observers. The variation within an observer (intra-observer variation) is assumed to be random; therefore, it does not cause bias in itself, but affects precision. In contrast, the variation between two observers (inter-observer variation) can introduce bias and make it difficult to interpret results.

Traditional static scoring of embryos has shown substantial inter-observer as well as intra-observer variability, which significantly impact the clinical success of morphology-based embryo selection (Matson, 1998; Arce et al., 2006; Baxter Bendus et al., 2006; Paternot et al., 2009; Ruiz de Assin et al., 2009). The high inter- and intra-observer variability found for the clinical decision of transfer, cryopreservation or rejection (Matson, 1998; Arce et al., 2006; Ruiz de Assin et al., 2009; Paternot et al., 2011) confirms this limitation. Knowledge about precision is therefore crucial when assessing the relevance of embryo evaluation parameters. Whereas intra-and inter-observer variability has been thoroughly investigated with regard to traditional static morphological parameters, there is limited knowledge regarding random errors and bias associated with time-lapse annotations. The present study was therefore conducted to assess the variability of time-lapse annotations.

Methods

Subjects

The study sample was drawn from patients attending the Fertility Clinic, Aarhus University Hospital, Denmark for IVF/ICSI treatment from February 2011 to June 2012. During this period, couples undergoing IVF/ICSI treatment were offered blastocyst culture to Day 6 and time-lapse monitoring as part of a cohort study evaluating parameters for embryo selection (manuscript in preparation). The cohort from whom the sample was obtained consisted of infertile patients aged <38 years without endometriosis. Patients were offered participation and their embryos included if ≥ 8 oocytes were retrieved. Twenty ($n = 20$) patients were randomly selected from the cohort. All embryos that displayed a normal fertilization (two pronuclei) were assessed. In total, 158 embryos representative for good prognosis patients were included in the present study.

Ethical approval

Written informed consent was obtained from all patients before inclusion. The Central Denmark Region Committees on Biomedical Research Ethics and the Danish Data Protection Agency approved the study.

Ovarian stimulation, oocyte retrieval and ICSI/IVF

Ovarian stimulation and oocyte retrieval were performed according to standard procedures as previously described by Kirkegaard et al. (2012). Following retrieval, the oocytes were fertilized using conventional IVF ($n = 20$) or ICSI ($n = 138$). After fertilization, ICSI embryos were immediately placed in the time-lapse incubator (EmbryoScope™). IVF embryos were cultured for $\sim$18 h in a conventional incubator. Consequently, no variables prior to first cleavage, including PN appearance, PN abuttal, PN breakdown, were evaluated for IVF fertilized embryos. Before IVF embryos were transferred to the EmbryoScope, adhering cumulus cells were removed to ensure optimal image acquisition. All embryos were cultured until arrest or removal at Day 6 (Kirkegaard et al., 2012).

Time-lapse recording and annotation

Images were recorded automatically in seven focal planes every 20 min (15 μm intervals, 1280 $\times$ 1024 pixels, 3 pixels per μm, monochrome, 8-bit $<$0.5 s per image using single 1 W red LED) and saved for evaluation at an external workstation (Embryo Viewer™). Time-lapse recordings of the 158 embryos were annotated manually at the external workstation. $t = 0$ was defined as time of fertilization. For ICSI embryos t_0 was defined as the time of injecting the sperm into the oocyte. For IVF embryos t_0 was defined as the time of adding the sperm to the dish. The time of fertilization was programmed into the EmbryoScope when the slide was loaded. The 18 h delay in transfer to the EmbryoScope was introduced when placing the IVF embryos in the time-lapse system. Time points refer to the exact time where an image was recorded and are reported as hours after fertilization. The observer annotated the time point for the first image obtained of the events being analyzed. Definitions of investigated parameters are shown in Table I. The time-point assessment was applied to the dynamic parameters: PN appearance, PN abuttal and PN breakdown, first cytokinesis, appearance and disappearance of first nucleus after first division, appearance and disappearance of second nucleus after first division, first–seventh division, final divisions, compaction, morula, early and full expanded blastocyst, start of hatching and fully hatched blastocyst. Two parameters (multi-nucleation and evenness at the 2-cell stage) were traditional morphological and assessed by binary values yes and no. The evaluation of evenness at the 2-cell stage was performed subjectively by eye at a time point where the two nuclei were visible. No measuring tools from the Embryo Viewer were used for the assessment of evenness.

Annotations were performed in a sequential order starting by the first image recording until arrest or removal at 140 h. If one observer did not annotate time points of specific events due to unfocused or missing images, or if the events were overlooked, these data were treated as missing data. The numbers of annotated embryos may therefore differ between the intra-/inter-observer assessments.

Three observers (A, B and C) with different levels of experience performed the time-lapse annotations. Observer C was a medical student with no prior experience in time-lapse annotations. Observer B was an embryologist with long-term experience in traditional morphology-based evaluation. Observer A was a PhD, and MD, graduate and had the most experience in time-lapse scorings. To avoid bias from the influence of others, the observers worked separately. One observer (C) did the assessment twice. At the reevaluation, the embryos were listed in a new randomized order to prevent recall bias. The observers were blinded to the assessment of the other observers and observer C was blinded to the assessment from the first evaluation. The observers were handed identical lists of embryo identification numbers before the assessments and an annotation scheme manual of the 25 specific morphologic and dynamic parameters. A training session was initiated by observer A for both observers to visually demonstrate the definitions of parameters. No additional training sessions or post-training evaluation took place.

Statistics

The inter-rater and intra-rater reliabilities were evaluated by the intra-class correlation coefficient (ICC) (Shrout and Fleiss, 1979). ICC is computed

Table I Dynamic and morphologic parameters included in the assessment.

Parameters	Definition	Assessment
PN appearance	One PN is visible	Time point
PN abuttal	No visible space between the two PN	
PN breakdown	PN are no longer visible	
First cytokinesis	Appearance of cleavage furrow or considerable elongation (>15%) at one-cell stage	
First nucleus appearance	Appearance of first and second nucleus at 2-cell stage	
Second nucleus appearance		
First nucleus disappearance	Disappearance of first and second nucleus at the 2-cell stage	
Second nucleus disappearance		
1–7 cleavage division	Complete separation of two daughter cells by a cytoplasm membrane	
Final division	The final cleavage division to ≥9 cells, before the initiation of compaction	
Compaction	The first fusion of two cell boundaries followed by a decrease in embryo diameter	
Morula	The compaction process is complete for all cells. All cell boundaries are unclear.	
Early blastocyst	The initiation of blastulation when a blastocoel is visible	
Full expanded blastocyst	The blastocoel cavity fills out ≥50% of the embryo and an inner cell mass is well defined	
Hatching blastocyst	The trophectoderm herniation through the zona pellucida	
Hatched blastocyst	The embryo has fully escaped the zona pellucida	
Blastocoel contractions	Contraction; the measured diameter is >10% smaller than the diameter at the previous time point	
Evenness	Evenness at 2-cell stage	Binary
Multinucleation	Multinucleate at 2-cell stage	

The time-point assessment was applied to the dynamic parameters. Assessment of blastocoel contractions was both assessed by time point and number. Assessment of binary values yes and no was applied to morphologic parameters. PN, pronuclei.

from variance estimates obtained by partitioning the variance into between-subjects variability and within-subjects variability. It provides a single value estimate that reflects agreement and consistency within assessments. The calculations were performed with the statistical package R (version 2.13.2), including the package various coefficients of inter-rater reliability and agreement (IRR) (version 0.84). ICC can be interpreted as follows: 0–0.2 indicates *poor* agreement: 0.3–0.4 indicates *fair* agreement; 0.5–0.6 indicates moderate agreement; 0.7–0.8 indicates *strong* agreement and >0.8 indicates *almost perfect* agreement.

Results

Inter-observer agreement

The inter-observer agreement of investigated parameters is shown in Table II. The variability of dynamic parameters between the three observers is shown in Figure 1A–H.

The agreement ranged from ICC 0.27 to 1. The average agreement was ICC 0.81 and median of ICC 0.87. The five parameters with highest correlations coefficients were timing of pronuclei breakdown, fully hatched blastocyst stage, disappearance of first nuclei after first division and appearance of first and second nuclei after first division. Least agreement was found for assessment of appearance of pronuclei, multi-nucleation (at the 2-cell stage), evenness of blastomeres (at the 2-cell stage), final divisions and number of collapses during blastocyst expansion.

Intra-observer agreement

The intra-observer agreement for investigated parameters is shown in Table II.

The agreement ranged from ICC 0.47 to 1. The average agreement was ICC 0.85 and median ICC of 0.92. The five highest correlations were found for timing of pronuclei breakdown, appearance of first and second nuclei after first division, fully hatched blastocyst stage and cleavage to 8 cells (t8). Least agreement was found for assessment of appearance of pronuclei, multi-nucleation (at the 2-cell stage), evenness of blastomeres (at the 2-cell stage), final divisions and numbers of collapses during blastocyst expansion.

No significant difference was found between inter- and intra-observer variability, expressed by ICC values.

Discussion

To the best of our knowledge, this study is the first to describe the inter- and intra-observer variability of time-lapse annotations. In general, the correlation was high for all of the investigated parameters. Results of inter-observer variability showed an average value ICC of 0.8 and the median value ICC of 0.9, which indicate strong and almost, perfect agreement, respectively. Highest correlation was found for the timing of pronuclei breakdown, fully hatched blastocyst, first nuclei disappearance (all ICC 1), followed by assessment of first and second nuclei appearance after first division (ICC 0.97–0.98). There was almost perfect agreement (ICC > 0.8) for all cleavage stages, with highest precision for first division (ICC 0.96); and strong agreement (ICC 0.7–0.8) for post-compaction parameters, with highest precision for the stage of morula. Only multi-nucleation and evenness of blastomeres at 2-cell stage showed fair agreement (ICC 0.3). Calculation of intra-observer variability produced similar or non-significantly higher agreement for most parameters.

Table II Inter-observer and intra-observer agreement for assessed parameters

Parameter	Inter ICC	*n*	95% CI	Intra ICC	*n*	95% CI
PN appearance	0.678	113	0.568–0.764	0.469	118	0.1–0.684
PN abuttal	0.773	90	0.692–0.838	0.685	108	0.569–0.775
PN breakdown	0.999	130	0.998–0.999	1	136	0.999–1
First cytokinesis	0.92	138	0.895–0.94	0.909	142	0.876–0.934
First nucleus appearance	0.984	90	0.977–0.989	0.982	123	0.974–0.987
Second nucleus appearance	0.967	78	0.953–0.978	0.965	117	0.948–0.976
First nucleus disappearance	0.989	86	0.985–0.993	0.966	115	0.951–0.976
Second nucleus disappearance	0.75	75	0.66–0.824	0.744	109	0.646–0.817
First cleavage division	0.958	134	0.945–0.969	0.953	150	0.935–0.965
Second cleavage division	0.885	133	0.85–0.913	0.954	145	0.937–0.967
Third cleavage division	0.809	141	0.756–0.853	0.87	145	0.824–0.905
Fourth cleavage division	0.863	125	0.821–0.897	0.929	138	0.902–0.949
Fifth cleavage division	0.868	123	0.826–0.901	0.899	129	0.86–0.928
Sixth cleavage division	0.906	99	0.872–0.933	0.923	118	0.889–0.946
Seventh cleavage division	0.9	68	0.852–0.935	0.975	93	0.962–0.983
Final division	0.599	82	0.336–0.757	0.708	74	0.573–0.805
Compaction	0.686	114	0.581–0.769	0.817	122	0.745–0.87
Morula	0.737	111	0.642–0.81	0.908	116	0.87–0.936
Early blastocyst	0.894	114	0.8–0.939	0.932	114	0.899–0.954
Full expanded blastocyst	0.913	116	0.869–0.941	0.937	118	0.91–0.956
Hatching blastocyst	0.872	75	0.816–0.914	0.968	84	0.952–0.979
Hatched blastocyst	0.99	11	0.972–0.997	0.973	12	0.912–0.992
Blastocoel contractions	0.653	75	0.536–0.752	0.771	114	0.685–0.836
Evenness	0.272	115	0.146–0.4	0.514	141	0.382–0.625
Multinucleation	0.268	153	0.168–0.372	0.472	155	0.339–0.586
Median ICC	0.87			0.92		
Mean ICC	0.81			0.85		

Note: inter-observer and intra-observer agreement expressed by ICC and 95% confidence interval (CI) of ICC values. N, number of embryos annotated by all three observers (inter-observer assessment) or by repeated annotations (intra-observer assessment).
ICC, intra-class correlation coefficient; PN, pronuclei.

Consistency is crucial to the validity of embryo scoring and selection. Previous research with time lapse has proposed predictive parameters of embryo viability and clinical pregnancy. Using manual annotation of cell divisions the timing and optimal duration of developmental events have been identified including: (i) time between fertilization and PN abuttal (Payne *et al.*, 1997), (ii) time between fertilization and PN breakdown (Azzarello *et al.*, 2012), (iii) appearance of nuclei after first cleavage (Lemmen *et al.*, 2008), (iv) duration of first cytokinesis (Wong *et al.*, 2010); (v) time between the first and second mitosis (Wong *et al.*, 2010; Meseguer *et al.*, 2011), (vi) synchronicity of the second and third mitosis (Wong *et al.*, 2010; Meseguer *et al.*, 2011), (vii) time to reach 5-cell stage (Meseguer *et al.*, 2011) and (viii) duration of third cleavage cycle (Hashimoto *et al.*, 2012). Moreover, direct cleavage from 1 to 3 cells along with unevenness and multi-nucleation at the 2-cell stage have been reported to display a strong negative correlation to pregnancy. These parameters have therefore been proposed in a hierarchical selection model as de-selection parameters (Meseguer *et al.*, 2011). A recent study has correlated post-compaction parameters to aneuploidy (Campbell *et al.*, 2013). Their findings may underline the importance of blastocyst evaluation and indicate the importance of late embryo quality markers in time-lapse evaluation. Conclusions on which parameters are predictive are however ambiguous and optimal time points for events may vary between clinics (Chamayou *et al.*, 2013). In principle, this could be the result of imprecise evaluation or different culture conditions, endpoints studied, embryo populations and probably false rejections of true differences (type II error). All of the time-lapse parameters suggested by the literature showed in our study high intra- and inter-observer correlation. This finding supports the interpretation that the lack of congruence across independent studies more likely are due to differences in the population of embryos studied, intervals between and quality of image recordings and outcome, than variability within the observers.

Some limitations to the time-lapse markers should be noticed. First, parameters where the optimal time range for distinguishing viable embryos from arresting embryos exceeds by either the observer variability or the interval between the recordings are clinically irrelevant.

The observer variability will depend on the consistency of the annotations and the latter will depend on the acquisition frequency. Mean time

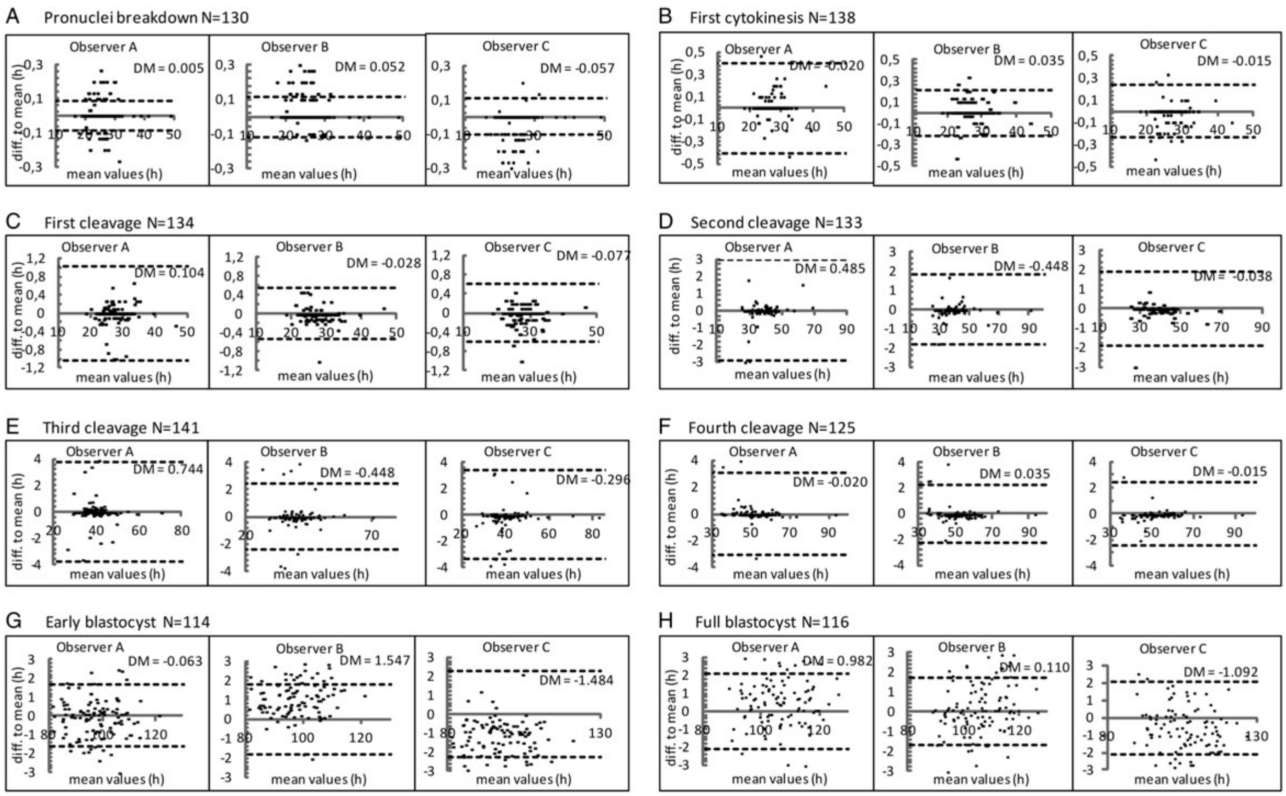

Figure 1 Bland–Altman plot of variability. The X-axis shows the mean of the three observers timing. The Y-axis is the difference of the observers' assessment against the mean (0); time expressed as parts of an hour. Number of investigated embryos (N). The mean of the differences (DM) and standard deviation of the differences (SD). Close clustering to mean equals high agreement. Plots of the variability of the annotations by the three observers for (**A**) breakdown of the pronuclei, (**B**) first cytokinesis, (**C**) first cleavage, (**D**) second cleavage, (**E**) third cleavage, (**F**) fourth cleavage, (**G**) early blastocyst and (**H**) full blastocyst. Observer C in the plot is the first score by observer C.

of completing first cytokinesis (14.3, ± 6.0 min) (Wong *et al.*, 2010) and the markers on early development under first cleavage cycle (∼<30 min) (Payne *et al.*, 1997; Lemmen *et al.*, 2008) were both suggested by studies that used systems with image acquisition every 1–10 min. It might therefore be suspected that these parameters would be difficult to assess using systems with longer intervals between the recordings. Our finding indicates that the assessment of first cytokinesis show almost perfect agreement and is easily recorded. This might be related to the morphologic change for this parameter (appearance of cleavage furrow) being both clear and over a short period of time. In contrast, some of the other events occurring during the first day of development are imprecisely evaluated, which might reflect that the change of morphology for these events is gradual and therefore more diffuse. In particular, we only found moderate to strong agreement (ICC 0.5–0.8) for the timing of pronuclei appearance and pronuclei abuttal. Secondly, some time points are annotated more precisely in normally developing embryos (Wong *et al.*, 2010). Results from this study showed better agreement for the assessment of the first nucleus disappearance than that of the second nucleus, which may reflect that embryos with rapid cleavage from 2 to 3 cells were part of the study sample. Furthermore, poor agreement for poor quality blastocysts and excellent agreement for good quality blastocysts have been reported in a comparison of manual annotations and the automatic tracking of cell divisions in time lapse, regarding the assessment of first cytokinesis and the mean time between first cleavage and second cleavage (Wong *et al.*, 2010). In our opinion, the findings reflect that assessments on abnormally short or long durations are characterized by more variability than assessments made on normal developing embryos. Therefore, when comparing and evaluating agreement between and within observers, the embryo population used must be recalled.

We demonstrated almost perfect agreement for the dynamic parameters, and fair agreement for the binary morphology-based parameters evaluated at the 2-cell stage, multi-nucleation and unevenness of the blastomeres. The fact that exact time point for evaluation of evenness was not defined for the observers; however, adds to this uncertainty. In principle, scoring where the observer has to make assessment in categories appears to be less precise than the analysis by the absolute timing (Paternot *et al.*, 2009). Similar difficulties in the assessment of multi-nucleation and unevenness have been reported using multilevel-images and 2D images, showing only moderate- to good inter-observer agreement (Arce *et al.*, 2006; Paternot *et al.*, 2009; Paternot *et al.*, 2011). In addition, we found more variability for seventh division (t8) compared with first division (t2). This might reflect that it is more difficult to count cells and to distinguish fragmentation, when cells are small and/or in layers.

Traditional static scoring of embryos has shown substantial inter-observer as well as intra-observer variability, which significantly impairs the quality control of morphology-based embryo selection (Matson, 1998; Arce et al., 2006; Baxter Bendus et al., 2006; Paternot et al., 2009; Ruiz de Assin et al., 2009). Distinct differences in the evaluated score, assessed images and the observer's training, complicate comparison between reproducibility assigned to static scoring and time-lapse annotations. Apart from control surveys, training sessions is one of the most efficient approaches to reduce inter-observer variations (Arce et al., 2006; Castilla et al., 2010; Paternot et al., 2011; Machtinger and Racowsky, 2013). In general, random errors are caused by mistakes and fluctuations in the readings and can be minimized by repeated annotations, whereas inter-observer variation is depending on both random and systematic errors (bias). To use observers with equal experience and background has been shown to be a successful method of limiting bias (Arce et al., 2006; Baxter Bendus et al., 2006; Ruiz de Assin et al., 2009; Paternot et al., 2011). Manual annotation in time lapse allows the observer to make repeated sequential observations on the same embryo. Furthermore, different observers can perform the assessment on recorded images. Taken together, this potentially allows for improved consistency and quality control of the embryo assessment. Currently, automated analysis in time lapse is limited beyond 4-cell stage (Wong et al., 2013) which indicates that potentially important embryo quality markers that occur later in development will require manual annotation for the time being.

It should be noted that the encouraging results from this study might reflect that the parameters were well defined and that the most experienced observer held the training session. We do, however suspect bias for some parameters, due to lack of experience in one observer. This concern arises from the reduced inter-observer agreement in comparison with the intra-observer agreement for the parameters: final divisions, compaction, morula and evenness of blastomeres (Table II). The results indicate that static morphology-based parameters remain rather subjective, and therefore require more experience for a precise categorization. A different group of observers with more similar background could potentially have produced higher correlation with regard to these particular markers. Although the influence of training was not investigated, our results emphasize the importance of ensuring well-defined parameters to equal scorings. Further investigations in multi-clinic design will be needed to confirm the external validity of our findings.

Conclusion

In conclusion, this study shows that dynamic predictors of viability can be assessed by extremely close agreement and are well defined even to novice users. Quality control of embryo assessments and selection may therefore be improved by a precise evaluation in time lapse.

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Authors' roles

K.K. and J.I. conceived the study and designed the project. L.S. drafted the manuscript. L.S. and K.K. did the acquisition and interpretation of data. K.K., U.B.K. and J.I. performed critical revision of the manuscript. All authors have given their final approval of present version to be published.

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Conflict of interest

None of the authors have any conflict of interest to declare.

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